

Noise-Aware Simulation and Statistical Eavesdropping Detection in BB84, Six-State, and E91 Quantum Key Distribution Protocols

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Abstract—The expected arrival of large scale quantum computers threatens the security of classical public key cryptosystems such as RSA and elliptic curve cryptography, whose security rests on computational hardness assumptions that Shor’s algorithm can defeat. Quantum Key Distribution (QKD) offers an alternative whose security is grounded in the laws of quantum mechanics rather than computational difficulty, allowing two parties to detect eavesdropping on the channel before a secret key is used. Prior simulation studies tend to evaluate single protocols, report a narrow set of metrics, and model an eavesdropper that attacks either every transmitted qubit or none. As a result, the way different QKD protocols expose a partial eavesdropper, and the cost of that detection in terms of usable key, remain poorly characterized. In this work, we present a unified Qiskit based study of 3 protocols, the 2 basis BB84 protocol, the 3 basis six state protocol, and the entanglement based E91 protocol, under matched ideal and noisy channel conditions. Beyond the standard metrics of Quantum Bit Error Rate (QBER), CHSH inequality violation, and secure key rate, we characterize eavesdropping detection statistically through the false positive rate and false negative rate, and we extend this analysis from full interception to partial intercept and resend, where the attacker targets only a fraction of the transmission. This yields, for each protocol, a tradeoff between detection performance and secure key rate as a function of attack strength and noise. Statistical detection of this kind was established for BB84 and recently extended to the six state protocol for prepare and measure QKD. We adopt those thresholds and provide the first false positive and false negative analysis of entanglement based CHSH detection, together with the first comparison of statistical detection across prepare and measure and entanglement based protocols within a single framework. The study establishes a reproducible environment that can be extended toward larger key lengths and real device noise. Across matched ideal channels, a partial attacker who intercepts up to 30% of the transmission evades detection in all 3 protocols while a positive secret key is still produced, and per session detection becomes reliable only near an interception fraction of 1/2, by which point the secure key rate has already collapsed. The six state protocol closes this stealth window earliest

and needs the fewest compared signals to suppress false alarms, BB84 is intermediate, and the entanglement based detector of E91 is the most evadable and the most fragile to channel noise, with false alarms rising sharply once the honest error rate exceeds about 8%. Under the measured `ibm_marrakesh` device noise, however, all 3 protocols operate honestly without false alarms.

Index Terms—Quantum Key Distribution, BB84, six-state protocol, E91, Qiskit, QBER, CHSH, secure key rate, eavesdropper detection, false positive rate, false negative rate

I. INTRODUCTION

Cryptographic systems have long protected sensitive communication by encoding plaintext into ciphertext that can only be recovered with the correct key. Classical cryptography achieves confidentiality, authenticity, and integrity through symmetric and asymmetric schemes such as the Advanced Encryption Standard (AES), the Data Encryption Standard (DES), the Rivest Shamir Adleman (RSA) algorithm, and Elliptic Curve Cryptography (ECC). The security of these schemes depends on the computational difficulty of mathematical problems such as integer factorization and the discrete logarithm. For decades, this assumption has been sufficient, because no classical algorithm can solve these problems efficiently at the key sizes used in practice.

The development of quantum computing changes this picture. Algorithms such as Shor’s algorithm and Grover’s algorithm can solve, or substantially accelerate the solution of, the problems on which classical public key cryptography depends. A sufficiently large quantum computer would therefore render much of the currently deployed asymmetric cryptography insecure. This threat is not limited to future communication. In a harvest now, decrypt later attack, an adversary records encrypted traffic today and decrypts it once a capable quantum computer becomes available, which exposes data that is

expected to remain confidential for many years. This prospect has motivated an urgent search for communication methods that remain secure against quantum adversaries.

Quantum Key Distribution (QKD) addresses this problem from a fundamentally different direction. Rather than relying on computational hardness, QKD derives its security from the laws of quantum mechanics, in particular the no cloning theorem and the measurement disturbance principle. Because an unknown quantum state cannot be copied, and because any measurement of a quantum state in the wrong basis disturbs it, an eavesdropper who attempts to intercept the quantum channel inevitably introduces detectable errors. The legitimate parties, traditionally called Alice and Bob, can therefore estimate the error rate across a portion of their exchanged bits and decide whether the channel is secure before deriving a final key. QKD does not transmit the message itself over the quantum channel. Instead, it establishes a shared secret key that is later used with a conventional encryption scheme, while an authenticated classical channel supports basis reconciliation and error checking.

To make the mechanism of QKD concrete, it is useful to recall how quantum information is represented and manipulated. The basic unit is the quantum bit, or qubit, whose state is a unit vector in a 2 dimensional complex space spanned by the computational basis states $|0\rangle$ and $|1\rangle$. Unlike a classical bit, a qubit may occupy a superposition $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with $|\alpha|^2 + |\beta|^2 = 1$. Measuring it in the computational basis returns 0 with probability $|\alpha|^2$ or 1 with probability $|\beta|^2$, and collapses the state onto whichever outcome occurs. Single qubit gates rotate this state, and 3 of them suffice for the protocols studied here. The Pauli X gate exchanges $|0\rangle$ and $|1\rangle$, acting as a bit flip. The Hadamard gate H maps the computational states to the equal superpositions $|+\rangle$ and $|-\rangle$, moving between the two conjugate bases that BB84 uses. The phase gate S adds a relative phase, which opens up a third measurement basis. These bases are what make the protocols work: a qubit gives a deterministic result when measured in the basis it was prepared in, but a uniformly random result in a conjugate basis, so an eavesdropper who guesses the wrong basis turns her mistake into observable errors. Two qubit gates such as the controlled NOT create entanglement, a joint state that cannot be factored into separate single qubit states and whose measurement outcomes stay correlated no matter how far apart the 2 parties are. Encoding bits into single qubit states, measuring them in randomly chosen bases, and testing the correlations of entangled pairs are the operations the three protocols are built from, and a circuit framework such as Qiskit expresses each one as a gate applied to a simulated qubit.

The first and most widely studied QKD protocol, BB84, was introduced by Bennett and Brassard in 1984 [1]. It encodes classical bits into the polarization states of single photons using 2 conjugate bases, the rectilinear basis at 0 and 90 degrees and the diagonal basis at 45 and 135 degrees. Alice prepares each qubit in a randomly chosen basis, Bob measures each qubit in a randomly chosen basis, and the two parties keep only the bits for which their bases agree, a step known as sifting or

basis reconciliation. The six state protocol generalizes BB84 by adding a third mutually unbiased basis, so that Alice and Bob choose among 3 bases rather than 2. The additional basis lowers the chance that an eavesdropper guesses the correct basis. This raises the disturbance an intercept and resend attack imposes, from 25% to 33%, and so improves the prospects for detection. The cost is a lower fraction of matching bases, and therefore a shorter sifted key. A complementary approach, the E91 protocol proposed by Ekert in 1991 [2], replaces prepare and measure with shared entanglement. A source distributes pairs of entangled qubits to Alice and Bob, who measure them along randomly chosen axes. The correlations between their outcomes are tested against the Clauser Horne Shimony Holt (CHSH) form of Bell's inequality, and a violation of this inequality certifies that the entanglement was genuine and that no eavesdropper has intercepted the qubits. These 3 protocols span the main design choices in discrete variable QKD, namely the number of preparation bases and the use of entanglement, which makes them a natural set for a controlled comparison.

Although these protocols are well understood theoretically, evaluating their behaviour under realistic conditions remains an active research problem. Physical QKD hardware is expensive and inaccessible to most researchers, which has made circuit level simulation an essential tool for studying protocol performance. Open source quantum computing frameworks such as Qiskit allow researchers to construct QKD circuits, introduce controlled noise through configurable channel models, and measure key performance indicators on classical hardware or cloud based simulators. However, existing simulation work shares three limitations. First, protocols are usually studied in isolation or, when compared, are reported only through coarse metrics such as the raw QBER. Second, the statistical analysis of eavesdropping detection through the false positive and false negative rates was first developed for BB84 and has recently been extended to the six state protocol, but it has never been applied to entanglement based detection, where the security check is a Bell CHSH test rather than an error rate. Third, although the partial intercept and resend attacker that targets only a fraction of the transmission has been studied for BB84 and the six state protocol, it has not been applied to entanglement based detection, and no work reports the resulting tradeoff between detection performance and secure key rate across prepare and measure and entanglement based protocols together.

In this work, we address these gaps through a simulation based study of 3 QKD protocols implemented in Qiskit, whose defining properties are summarized in Table I. The contributions of this study are summarized as follows.

- 1) We implement the BB84, six state, and E91 protocols within a single Qiskit based framework, covering qubit preparation, basis reconciliation, error reconciliation, and privacy amplification, so that the 3 protocols can be compared on a common footing under matched ideal, synthetic noise, and real device noise conditions.
- 2) Beyond the standard metrics of QBER, CHSH violation, and secure key rate, we characterize eavesdropping

detection statistically through the false positive and false negative rates. Statistical detection for prepare and measure protocols was established for BB84 by Lee et al. [3] and for the six state protocol by Fiorini et al. [4]; we adopt these thresholds and, for the first time, extend the false positive and false negative analysis to the entanglement based CHSH detection of E91, placing prepare and measure and entanglement based detection on a common footing.

- 3) We model a partial intercept and resend attacker that targets a tunable fraction of the transmission, and we determine for each protocol the attack strength at which detection becomes unreliable, producing a tradeoff between detection performance and secure key rate as a function of attack strength and noise.

The rest of this paper is organized as follows. Section II reviews the related work on QKD simulation, implementation, and security analysis. Section III provides the necessary background on the BB84, six state, and E91 protocols and the simulation environment. Section IV presents the simulation pipeline and the performance metrics used in the evaluation. Section V presents the experimental results and discussion. Finally, Section VI concludes the paper and outlines directions for future work.

TABLE I
SUMMARY OF THE THREE QKD PROTOCOLS COMPARED IN THIS WORK

Property	BB84	Six-State	E91
Type	Prepare-and-measure		Entanglement
Bases / settings	Z, X	Z, X, Y	Random axes
Sifting ratio	$\approx 1/2$	$\approx 1/3$	Matched axes
Full-attack signal	25% QBER	33% QBER	$S \rightarrow 1.41$
Detection signal	QBER	QBER	CHSH S
Detection threshold	0.135	0.167	$S = 2$

II. RELATED WORK

Quantum Key Distribution has attracted sustained research attention as classical cryptography faces the prospect of quantum attacks. This section reviews prior work on the simulation, implementation, and security analysis of QKD protocols, with a focus on BB84 and E91. We organize the discussion into 4 themes. We first consider studies that implement both protocols together, then studies that focus on a single protocol, then work on security analysis and attack models, and finally simulation frameworks that support QKD research. Table II summarizes how these studies compare against the present work across a set of representative features.

A. Combined BB84 and E91 Implementations

A small number of studies evaluate BB84 and E91 together, which makes them the closest precedents for the present work. Ain et al. [5] proposed an enhanced BB84 scheme that uses

9, 12, and 16 qubits across 2 and 3 bases, including computational, Hadamard, and diagonal bases, in order to strengthen security against brute force and intercept and resend attacks. Their study reports that increasing the qubit count makes it progressively harder for an eavesdropper to correctly guess both the transmitted state and the measurement basis, which raises the probability of detection. They further implemented the E91 protocol on the IBM Quantum Experience platform and introduced a reverse gate configuration that collapses the computational basis state from one form to another and attaches a misleading phase angle, so that an intercepted state yields deceptive information to the attacker. To handle the additional noise introduced by the 2 qubit gates that entanglement requires, they applied error mitigation during the final measurement. Their results, gathered through Qiskit simulations with 1024 shots, show fluctuations in the measured probabilities that reveal eavesdropping and confirm that the 16 qubit configuration provides the lowest information leakage. Dhakal et al. [6] provided a further direct precedent, evaluating BB84, E91, and an enhanced BB84 variant in Qiskit under several noise models, including depolarising, thermal relaxation, Pauli, amplitude damping, and phase damping channels, across key lengths ranging from 200 to 3000 bits. Their study used the QBER and the CHSH violation as the primary evaluation metrics and emphasized the trade off between security and efficiency across the 3 protocols.

B. Single Protocol Simulations

Several studies focus on a single protocol. Saeed et al. [7] presented one of the first published quantum circuits for BB84 in Qiskit, simulating the protocol both with and without an eavesdropper on a local simulator and on an IBM cloud simulator. They established the presence of an eavesdropper by observing how the probability distributions of the measured outcomes shift when interception occurs, and they discard and regenerate the key when this shift indicates a compromised exchange. On the entanglement based side, Kadir and Hossain [8] simulated E91 on the Qiskit AerSimulator, with a third party preparing singlet states and Alice and Bob each choosing randomly among 3 measurement bases. They reported a CHSH parameter of 2.47, which exceeds the classical bound of 2 and confirms a genuine Bell violation, and showed that the key generation rate rises with the number of singlet states, reaching about 12 bits per second for 500 pairs. Mariani et al. [9] implemented the complete E91 protocol on the IBM Quantum platform using a depolarizing noise channel and incorporated the classical post processing stages of error correction and privacy amplification. In their channel model the depolarizing strength maps to an expected QBER of $1/2$ its value, and they concentrated their analysis on the region around a 10% error rate, near the level at which key distillation becomes marginal. By modeling the protocol end to end, including these post processing steps, they positioned their implementation as a reusable tool for studying not only point to point QKD but also more complex systems that rely on QKD for security. Badiezedegan [15] described a single

TABLE II
COMPARISON OF RELATED WORK ON QKD SIMULATION AND ANALYSIS

Feature	[5]	[6]	[7]	[8]	[9]	[3]	[10]	[11]	[12]	[13]	[4]	[14]	Our Work
Implements BB84	✓	✓	✓	✗	✗	✓	✗	✗	✗	✓	✗	✓	✓
Six-state protocol	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✓	✗	✓
Implements E91	✓	✓	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓
Noise Modeling	✓	✓	✗	✗	✓	✓	✗	✗	✓	✓	✓	✓	✓
QBER / CHSH	✓	✓	✓	✓	✓	✓	✓	✓	✗	✓	✓	✓	✓
Secure Key Rate	✗	✓	✗	✓	✓	✗	✓	✓	✗	✗	✗	✗	✓
FPR / FNR Detection	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✓	✓	✓
FPR / FNR for E91	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Partial Intercept Attack	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓
Multi-protocol Comparison	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓

protocol BB84 implementation that models qubits as explicit state vectors and carries the protocol through the full classical pipeline, including parity based block error reconciliation and binary entropy based privacy amplification. While that work is educational and treats BB84 alone, its post processing stages provide a concrete reference for the reconciliation and privacy amplification steps required to compute a realistic secure key rate.

C. Security Analysis and Attack Models

A further body of work analyzes the security of QKD protocols and the attacks to which they are exposed. Lee et al. [3] provided the most directly relevant analysis, studying eavesdropping detection in BB84 as a statistical decision problem. Using Hoeffding’s inequality they derived upper bounds on the false positive rate and false negative rate of detection, where an eavesdropper is declared whenever the measured QBER reaches a threshold. By choosing the threshold that minimizes the combined bound, an optimal value near a QBER of 0.135, they showed that comparing about 300 qubits keeps both error rates below roughly 0.009%. They also proposed a grouped BB84 variant for rapidly varying channels. Their analysis is confined to BB84 and to an attacker that intercepts every qubit, and they explicitly leave 3 problems open, namely the evaluation of exact rather than upper bound error rates, the case of an attacker that intercepts only part of the transmission, and the resulting tradeoff between detection performance and secret key rate.

A closely related line of work by Fiorini, Garroppo, Pagano, and collaborators develops statistical eavesdropper detection

for prepare and measure protocols under a partial intercept and resend attack, and is the closest precedent to the prepare and measure portion of the present study. They first studied the estimation of the interception density in BB84 using a noisy Qiskit simulator [13], then derived two optimal QBER thresholds for the six state protocol that minimize the false positive and false negative rates, one from upper Hoeffding bounds and one from a limiting probability density approach, validated on backends inspired by real IBM devices [4]. A subsequent work proposed a BB84 threshold in which the two communicating parties agree on the maximum fraction of key bits the attacker may correctly intercept, removing the dependence on an estimated interception density and setting the threshold from the system noise and the application’s security requirements [14]. We adopt the threshold construction of this line for BB84 and the six state protocol and extend the analysis to entanglement based detection, which it does not consider.

The statistical disturbance signature that this detection exploits for the six state protocol was established by Bechmann-Pasquinucci and Gisin [10], who classified the incoherent and coherent eavesdropping strategies available against the 3 basis protocol and showed that the additional mutually unbiased basis raises the disturbance an intercept and resend attacker imposes relative to BB84. Their analysis provides the theoretical basis for the larger error signal that makes six state detection more sensitive, which the present work measures empirically as a false positive and false negative rate. Complementing this statistical view of prepare and measure detection, Singh et al. [11] analyzed the relationship between

the Bell-CHSH violation and the secure key rate of entanglement based QKD through a geometric representation of 2 qubit correlations. They showed that a violation of the CHSH inequality is a necessary boundary for distilling a secret key, though it is not on its own sufficient, since states exist that violate the inequality yet yield no usable key, and others that yield a key only after local filtering. Their result grounds the use of the classical bound of 2 as the security relevant decision boundary for the CHSH based detection of E91 in the present study, and it clarifies that crossing this boundary marks the point beyond which the entanglement can no longer certify a secure key. Fei et al. [16] demonstrated a quantum man in the middle attack that targets the calibration process of a BB84 system rather than the key exchange itself. By intercepting calibration signals and resending faked ones, the attacker induces a detector efficiency mismatch that opens a side channel, and the authors proposed 2 countermeasures to remove this vulnerability. These studies highlight that the security of QKD in practice depends not only on the idealized protocol but also on the physical implementation and its supporting procedures, which is relevant to any simulation that aims to model realistic conditions.

D. Simulation Frameworks

Beyond protocol specific work, several frameworks support QKD research at the network and circuit level. DiAdamo et al. [12] introduced QuNetSim, a Python framework for simulating quantum networks at the network layer that incorporates many known quantum networking protocols and allows users to build and test protocols, including QKD, over a range of network configurations. Li et al. [17] extended the Quantum Toolbox in Python with pulse level simulation of noisy circuits, enabling the modeling of environment induced decoherence and control pulse noise that is relevant to characterizing protocol behaviour on noisy intermediate scale quantum devices. At the system level, Zeng et al. [18] examined how QKD and post quantum cryptography can be combined into hybrid schemes, weighing the advantages and limitations of each approach and motivating the broader context in which QKD protocols are expected to be deployed.

E. Research Gap

As summarized in Table II, prior work has progressively built the statistical detection picture for prepare and measure QKD but has left entanglement based detection and the cross protocol comparison untouched. Lee et al. [3] cast eavesdropping detection in BB84 as a statistical decision problem and bounded the false positive and false negative rates under a full interception attack. Fiorini et al. extended this line to a realistic partial intercept and resend attacker, first estimating the interception density in BB84 [13], then deriving optimal false positive and false negative thresholds for the six state protocol under channel noise [4], and then proposing a refined BB84 threshold independent of the attacker's interception density [14]. These works are confined to prepare and measure protocols and to the QBER as the detection signal.

The theoretical analysis of entanglement based security [11] characterizes the CHSH violation as a necessary boundary for a secret key but does not cast it as an empirical detection test with measurable error rates. No prior work reports the false positive and false negative rates of the entanglement based CHSH detector, places prepare and measure and entanglement based detection within a single framework under a matched partial attacker, or characterizes the resulting tradeoff between detection performance and secure key rate across the three protocols. The present work addresses these gaps: we adopt the established BB84 and six state thresholds, extend the false positive and false negative analysis to the CHSH detection of E91 for the first time, and compare all three protocols on the detection versus key rate tradeoff as the attack strength varies.

III. SYSTEM MODEL

This section describes the three protocols and the threat and noise models used throughout the study. All protocols share the same setting. Two legitimate parties, Alice and Bob, are connected by a quantum channel that carries qubits and by an authenticated classical channel that carries reconciliation messages. An eavesdropper, Eve, has full access to the quantum channel and may intercept qubits, but cannot tamper with the authenticated classical channel. This common setting is summarized in Fig. 1. Each protocol is implemented as a quantum circuit in Qiskit and executed on the Aer simulator, with noise introduced as described in Section III-E.

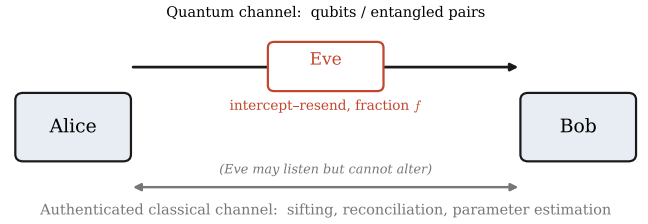


Fig. 1. System and threat model. Alice and Bob share a quantum channel carrying qubits or entangled pairs and an authenticated classical channel for sifting, reconciliation, and parameter estimation. Eve runs an intercept-and-resend attack on a fraction f of the quantum channel and may read but not alter the classical channel.

A. BB84 Protocol

In BB84, Alice generates 2 independent random bit strings, one that sets the key bit and one that selects the preparation basis. For each position she encodes the key bit in either the rectilinear basis $Z = \{|0\rangle, |1\rangle\}$ or the diagonal basis $X = \{|+\rangle, |-\rangle\}$. A bit value of one is encoded by an X gate, and a choice of the diagonal basis is realized by a Hadamard gate H , so that the prepared state is one of the 4 signal states $|0\rangle, |1\rangle, |+\rangle, |-\rangle$, as shown in the BB84 circuit of Fig. 2. These 4 states form 2 conjugate, or mutually unbiased, bases, meaning that a state prepared in one basis yields a completely random outcome when measured in the other. The protocol proceeds in 2 phases. In the quantum

phase, Alice transmits the prepared qubits one by one over the quantum channel. Bob selects an independent random basis for each qubit, applying a Hadamard gate before measurement when he chooses the diagonal basis and measuring in the computational basis otherwise. In the classical phase, carried out over the authenticated public channel, Alice and Bob announce their basis choices, keep only the positions where the bases agree, and discard the rest. This step, called sifting or basis reconciliation, yields a shared sifted key. Because the two bases are chosen independently and uniformly, they agree on about 1/2 of the positions, so the expected sifting ratio is 1/2.

The security of BB84 rests on 2 principles of quantum mechanics. The no cloning theorem forbids an eavesdropper from copying an unknown qubit, so she cannot keep a copy and forward the original untouched. The measurement disturbance principle then ensures that any measurement she performs in the wrong basis randomizes the state she resends. As a result, an eavesdropper who does not know the preparation basis inevitably introduces discrepancies between Alice's and Bob's sifted bits, which the parties detect by comparing a random sample and estimating the error rate. If that rate is low enough, the remaining sifted bits are passed to error reconciliation and privacy amplification to produce the final secret key; if it is too high, the exchange is aborted.

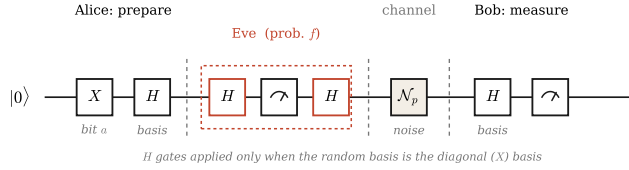


Fig. 2. BB84 circuit for a single qubit. Alice encodes a random bit with X and selects a basis with H ; Eve, applied with probability f , intercepts, measures, and resends; the channel applies depolarizing noise $\mathcal{N}_p$; Bob rotates to his random basis and measures. The H gates act only for the diagonal (X) basis.

B. Six-State Protocol

The six state protocol generalizes BB84 by adding the third mutually unbiased basis $Y = \{|+i\rangle, |-i\rangle\}$, whose states are $|\pm i\rangle = (|0\rangle \pm i|1\rangle)/\sqrt{2}$, so that Alice and Bob each select uniformly among the 3 bases Z , X , and Y . These 3 bases are the eigenbases of the Pauli operators and are pairwise conjugate, which makes them the complete set of mutually unbiased bases available to a single qubit; Fig. 3 contrasts them with the two bases of BB84. Preparation in the Y basis is obtained from the computational basis by the gate sequence SH , and measurement in the Y basis by the inverse sequence $HS^\dagger$, where S is the phase gate. Preparation and measurement in the Z and X bases are identical to BB84. The full circuit is shown in Fig. 4. All remaining steps, sifting and classical post processing, are unchanged. Two consequences follow from the additional basis. First, because Alice and Bob now choose independently among 3 bases, they agree on only about 1/3 of the positions, so the sifting ratio falls from 1/2 to 1/3 and the

protocol yields fewer key bits per transmitted qubit. Second, an eavesdropper performing an intercept and resend attack must guess among 3 bases rather than 2, so she chooses correctly less often and disturbs the state more frequently when she is wrong. The expected error she imposes on the sifted key therefore rises from 25% for BB84 to about 33 % for the six state protocol. The six state protocol thus trades key yield for a stronger and more visible response to eavesdropping, and this explicit tension between efficiency and detectability is the property that motivates its inclusion in this comparison.

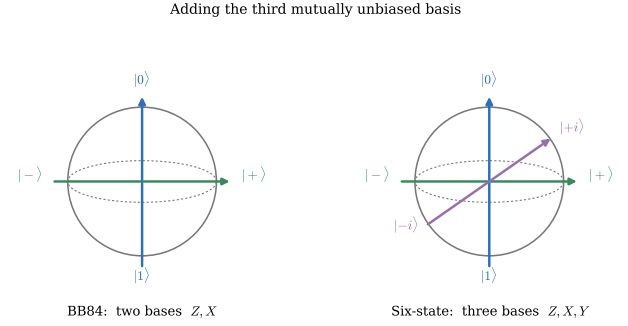


Fig. 3. Measurement bases as axes on the Bloch sphere. BB84 uses two mutually unbiased bases, Z and X ; the six-state protocol adds the third, Y . The extra basis raises the intercept-and-resend disturbance from 25% to 33% at the cost of a smaller sifted key.

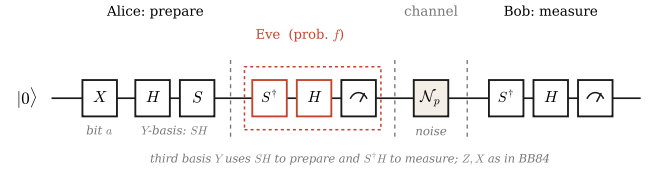


Fig. 4. Six-state circuit for a single qubit. The third basis Y is prepared with SH and measured with $S^\dagger H$, while the Z and X bases are as in BB84. Eve and the channel act as in Fig. 2.

C. E91 Protocol

The E91 protocol replaces prepared single qubits with shared entanglement and derives its security from a Bell test rather than from basis disturbance. For each round a source prepares a maximally entangled Bell pair. A Hadamard gate on the first qubit, followed by a controlled NOT gate onto the second, produces the state $(|00\rangle + |11\rangle)/\sqrt{2}$. The source sends one qubit to Alice and one to Bob, as shown in Fig. 5. Because the pair is entangled, a measurement on one qubit instantaneously fixes the correlated outcome of the other, no matter how far apart the 2 parties are. Alice and Bob each measure their qubit along an independently and randomly chosen axis, implemented as a R_y rotation that selects the measurement angle followed by a computational basis measurement. Their chosen angles fall into 2 groups. When their axes are aligned, their outcomes are strongly correlated and the corresponding bits are kept to form the shared key, which plays the role that the matched basis positions play in BB84. When their axes

differ in the prescribed way, the outcomes are used not for key generation but to estimate the CHSH quantity

$$S = E(a, b) - E(a, b') + E(a', b) + E(a', b'), \quad (1)$$

where a, a' and b, b' are Alice's and Bob's alternative measurement axes and $E(\cdot, \cdot)$ is the correlation between their outcomes for a given pair of axes, defined as the average of the product of the ± 1 outcomes. For a genuinely entangled pair measured at the optimal angles, S can reach the Tsirelson bound $2\sqrt{2} \approx 2.83$, whereas any classical or local hidden variable description obeys the bound $S \leq 2$. A measured value of S above 2 therefore certifies that genuine entanglement survived transmission. An eavesdropper who intercepts and measures a qubit destroys part of this entanglement, which weakens the correlations and drives S back toward the classical bound, so the collapse of S toward two is the signature that the channel has been disturbed. This entanglement based check is appealing because it tests the physical resource that secures the key directly, rather than inferring an attack from an error rate alone.

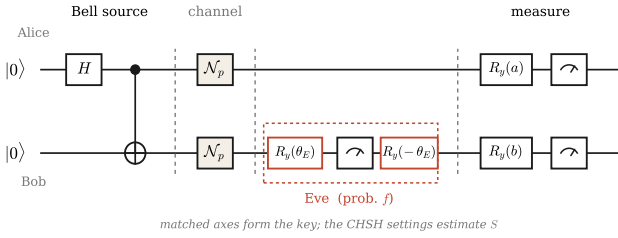


Fig. 5. E91 circuit. A Bell pair is created with H and CNOT and shared between Alice and Bob; the channel applies $\mathcal{N}_p$; Eve, with probability f , measures Bob's qubit along a random axis and resends; Alice and Bob measure along randomly chosen axes $R_y(a)$ and $R_y(b)$. Matched axes form the key and the remaining settings estimate S .

D. Threat Model: Partial Intercept and Resend

We adopt the intercept and resend attack as the adversary model. For an attacked qubit, Eve measures in a randomly chosen basis and forwards a freshly prepared qubit consistent with her outcome, which disturbs the state whenever her basis differs from the preparation basis. Following the partial intercept and resend model introduced for BB84 by Fiorini et al. [13] and applied by them to the six state protocol [4], we parameterize the attack by an interception fraction $f \in [0, 1]$, so that Eve attacks each qubit, or each entangled pair, independently with probability f . The case $f = 1$ recovers the full attack studied in prior work, while intermediate values model an adversary that limits her interference to remain below the detection threshold. We apply this attacker for the first time to the entanglement based E91 protocol, so that a single adversary model can be compared across all three protocols. For E91, interception of one member of a pair collapses the entanglement and reduces the measured CHSH value.

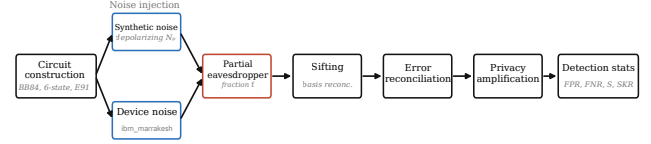


Fig. 6. End-to-end simulation pipeline. Each run proceeds from circuit construction, through one of the two mutually exclusive noise regimes (synthetic depolarizing or the `ibm_marrakesh` device model), the partial intercept-and-resend eavesdropper acting on a fraction f of the transmission, and the classical post-processing stages, to the final computation of the error indicators and detection statistics.

E. Noise Model

Two noise regimes are considered. The first is a synthetic per qubit depolarizing channel of strength p , applied uniformly to every transmitted qubit so that the modeled disturbance does not depend on which gates a particular qubit happened to receive. This regime provides a clean, controllable axis along which to vary the channel quality. The second is a device noise model extracted from the `ibm_marrakesh` backend. Unlike the uniform depolarizing channel, this model reproduces the heterogeneous error sources of real hardware, including per gate error rates, thermal relaxation characterized by the qubit T_1 and T_2 times, and readout errors that vary from qubit to qubit. Testing against these physical realities matters because they do not behave as a single tunable parameter; instead they natively inflate the baseline QBER and degrade the baseline CHSH value S even when no eavesdropper is present. Because honest sessions therefore begin closer to the abort condition, the device model inherently shrinks the stealth zone, the band of attack strengths an eavesdropper can occupy before the protocol is forced to abort, and so provides a stricter and more realistic test of detection than synthetic noise alone. The 2 regimes are treated as mutually exclusive, so that any reported error or detection statistic can be attributed to a single, clearly specified noise source.

F. Classical Post-Processing

For BB84 and the six state protocol, the sifted key is processed in 3 stages. Parameter estimation reveals a random subset of the sifted bits to measure the QBER, error reconciliation corrects the surviving mismatches using parity based block correction, and privacy amplification compresses the reconciled key to remove the information potentially leaked to Eve, with the compression length set from the binary entropy of the estimated QBER and the number of disclosed parity bits. For E91, the correlated matched basis outcomes furnish the key, which is then processed through the same reconciliation and privacy amplification stages, while the unmatched settings supply the CHSH estimate.

IV. PERFORMANCE METRICS

The end to end simulation pipeline is summarized in Fig. 6. Each run begins with circuit construction for the chosen

protocol, after which noise is injected from one of the two regimes of Section III-E, either the synthetic depolarizing channel or the `ibm_marrakesh` device model. The partial intercept and resend eavesdropper of Section III-D then acts on a fraction f of the transmission. The surviving signals pass through sifting, error reconciliation, and privacy amplification, and the run concludes by computing the error indicators and detection statistics, namely the QBER or CHSH value, the secure key rate, and the false positive and false negative rates.

We evaluate the three protocols using 2 groups of metrics. The first group contains standard quantities that validate the implementation and place it alongside prior work. The second group contains the detection statistics that form the contribution of this study.

A. Basic Metrics

Quantum Bit Error Rate (QBER). For BB84 and the six state protocol, the QBER is the fraction of positions in the sifted key where Alice's and Bob's bits disagree. It captures the combined effect of channel noise and eavesdropping and is the primary error indicator for prepare and measure protocols.

CHSH parameter. For E91, the CHSH value S of (1) plays the analogous role. A value above the classical bound of 2 certifies entanglement, and the amount by which S exceeds 2 measures the quality of the correlations that secure the key.

Secure key rate. For all three protocols, the secure key rate is the number of secret key bits produced per transmitted qubit. Following the asymptotic Shor and Preskill expression, the secret fraction is taken as $r = \max(0, 1 - 2H_2(Q))$, where Q is the QBER and H_2 is the binary entropy function, and the secure key rate is the product of this fraction and the sifting ratio. The rate falls to 0 once the QBER reaches the threshold near 11%, beyond which no secret key can be distilled.

Sifting efficiency. The sifting ratio is the fraction of transmitted qubits retained after basis reconciliation. It is about 1/2 for BB84 and about 1/3 for the six state protocol, and it quantifies the key yield cost of adding a third basis.

B. Novel Metrics: Statistical Detection

False positive rate (FPR). Eavesdropping is declared when the measured error indicator crosses a decision threshold, the QBER for BB84 and the six state protocol, and a corresponding threshold on S for E91. The false positive rate is the probability that this test declares an eavesdropper when none is present and only channel noise is acting. A high FPR means a protocol aborts honest sessions unnecessarily.

False negative rate (FNR). The false negative rate is the probability that the test fails to declare an eavesdropper when one is in fact present. A high FNR means an attacker passes undetected. The FPR and FNR together characterize the reliability of eavesdropping detection, and we estimate both empirically from repeated simulation runs rather than from analytic bounds. Fig. 7 shows how both error rates arise from the overlap of the honest and eavesdropper distributions at the decision threshold. For BB84 and the six state protocol the statistical detection framework follows Lee et al. [3] and

Fiorini et al. [4]; the contribution here is to report the same false positive and false negative rates for the entanglement based CHSH detection of E91, which has not previously been analyzed in these terms, and to compare all three on a common footing.

Decision thresholds. Two different limits involve the QBER, and we keep them strictly separate. The first is the secure key rate limit near a QBER of 11%, the asymptotic Shor and Preskill bound beyond which the distilled secret fraction reaches 0 and no key can be safely produced. This limit only dictates whether a key can still be distilled; it does not raise an alarm. The second, used for detection, is the statistical threshold that actually triggers the eavesdropper alarm, and it is a separate quantity. We initially explored calibrating the detection thresholds dynamically from each channel's baseline honest noise profile, placing the alarm a fixed margin above the observed honest level in order to suppress false alarms on noisy channels. We ultimately discarded this adaptive scheme in favor of fixed, literature grounded limits, so that the reported error rates are measured against rigorous and standardized security bounds rather than boundaries fitted to our own data. For BB84 we adopt the optimal QBER threshold of Lee et al. [3], who place the decision boundary that minimizes the combined Hoeffding bound on the 2 error rates at a QBER of 0.135. For the six state protocol we adopt the Hoeffding balanced threshold derived by Fiorini et al. [4]. When the QBER is estimated from n compared bits, Hoeffding's inequality bounds the two detection errors by

$$\text{FPR} \leq e^{-2n(\tau - q_0)^2}, \quad \text{FNR} \leq e^{-2n(D - \tau)^2}, \quad (2)$$

where τ is the decision threshold, q_0 is the honest error rate, and D is the expected disturbance under a full intercept and resend attack. Equalizing the two exponents places the threshold at the midpoint $\tau = (q_0 + D)/2$. For BB84, with $D = 1/4$, this coincides with the optimized value of Lee et al. [3]. For the six state protocol the full attack disturbance is larger, $D = 1/3$, giving $\tau = 1/6 \approx 0.167$ with a noiseless reference $q_0 = 0$, consistent with Fiorini et al. We use these established thresholds for the prepare and measure protocols and contribute the corresponding analysis for the entanglement based detector of E91, for which no statistical detection threshold has previously been reported. For E91 the decision boundary is the CHSH classical bound of 2.0, the value used to separate secure from compromised channels in prior entanglement based studies [2], [6], [8]. A Bell violation is a necessary condition for distilling a secret key [11], so a measured S that falls to 2 signals that the entanglement, and with it the security of the channel, can no longer be certified.

Detection resource cost. For a target FPR and FNR, we report the number of compared qubits, or entangled pairs, that each protocol requires to reach that target, which quantifies the quantum resources detection consumes.

Detection versus key rate tradeoff. Finally, by sweeping the interception fraction f of Section III-D, we report for each protocol the joint behaviour of detection performance and secure key rate, and we identify the attack strength at which

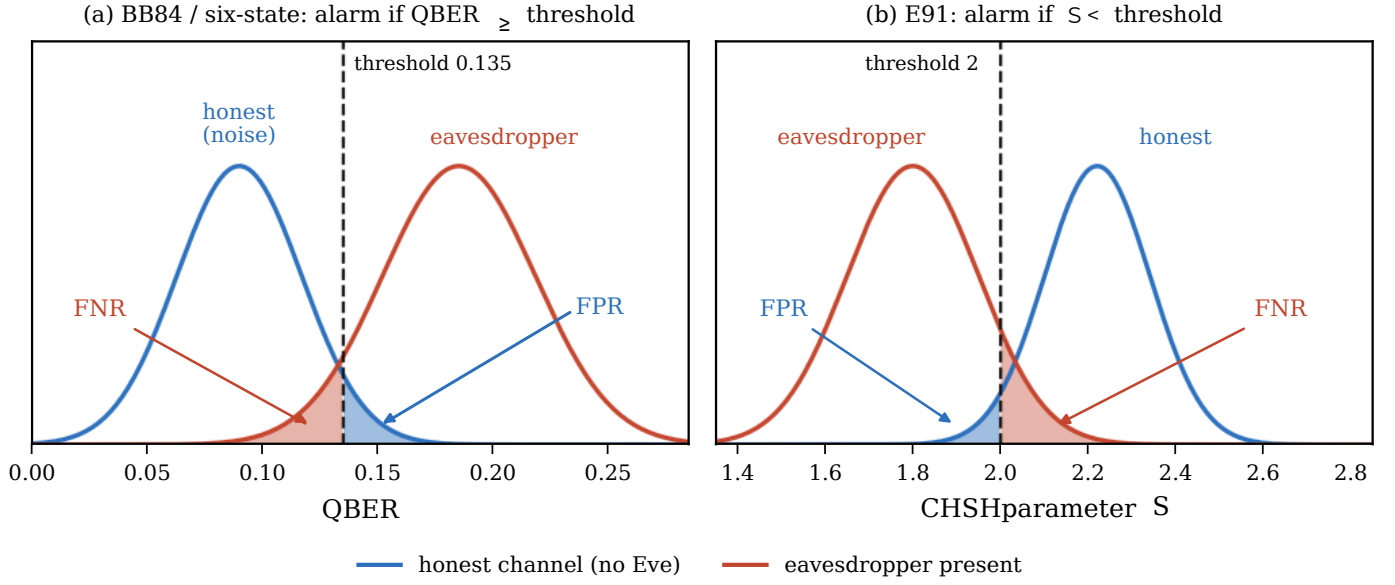


Fig. 7. Statistical definition of the detection errors. An alarm is declared when the error indicator crosses a fixed threshold. (a) For BB84 and the six-state protocol the alarm fires when the QBER reaches the threshold, so the false positive rate is the honest-channel mass above it and the false negative rate is the eavesdropper mass below it. (b) For E91 the alarm fires when the CHSH value S falls below the classical bound $S = 2$, mirroring the roles. Both rates are estimated empirically by Monte Carlo.

detection becomes unreliable. This tradeoff is the central result the framework is designed to produce.

V. RESULTS AND DISCUSSION

Unless otherwise stated, every synthetic data point is the mean over 50 independent Monte Carlo trials, using 1000 transmitted qubits per trial for BB84 and the six state protocol and 8192 entangled pairs per trial for E91; the device noise points use 30 trials, with 200 qubits or 1024 pairs per trial. The interception fraction f is swept from 0 to 1 in steps of 0.1 for the synthetic channels and in steps of 0.25 for the device model. We first validate the implementation against known analytic limits, then present the detection signal as a function of attack strength, the stealth window and the resulting tradeoff against secure key rate, the robustness of each detector to channel noise, the behaviour under real device noise, and finally the resource cost of detection.

A. Implementation Validation

On an ideal channel with no eavesdropper, the 3 protocols reproduce their textbook limits, which confirms that the circuits and the post processing are correct. BB84 yields a QBER of 0.000 and a secure key rate of 0.498, matching the expected 1/2 sifting ratio, and the six state protocol yields a QBER of 0.000 and a secure key rate of 0.333, matching its 1/3 sifting ratio. For E91 the honest CHSH value is $S = 2.823$, within statistical error of the Tsirelson bound $2\sqrt{2} \approx 2.83$, and its secure key rate is 0.223, consistent with the 2/9 setting combinations used for key generation. Under a full intercept and resend attack ($f = 1$) the disturbance signatures also match theory: the BB84 QBER rises to 0.253 and the six state QBER to 0.338, close to the predicted 25 and 33%, while the

E91 CHSH value falls to $S = 1.415 \approx \sqrt{2}$, well inside the classical region. In every case the secure key rate is 0 under the full attack. These anchor values appear as the endpoints of Fig. 8.

B. Detection Signal Versus Attack Strength

Fig. 8 shows how each protocol's error indicator responds as the interception fraction grows on an otherwise ideal channel. The QBER of BB84 and the six state protocol rises monotonically with f , and the CHSH value of E91 falls monotonically, so each indicator moves steadily from its honest value toward its full attack value. The rate of change differs across protocols. The six state QBER rises fastest, reaching its detection threshold of 0.167 between $f = 0.4$ and $f = 0.5$, because its 3 basis measurement makes Eve guess wrong more often. The BB84 QBER reaches its lower threshold of 0.135 between $f = 0.5$ and $f = 0.6$, and the E91 CHSH value crosses the classical bound $S = 2$ between $f = 0.5$ and $f = 0.6$. The important consequence is that for all 3 protocols the indicator does not reach its decision boundary until roughly 1/2 of the transmission is intercepted, which already suggests that a weaker attacker can remain below the alarm.

C. Stealth Window and Detection Versus Key Rate Tradeoff

Fig. 9 makes this tension explicit by plotting, against the same attack fraction, the false negative rate of detection in the upper panel and the secure key rate in the lower panel. Three regimes are visible. For $f \leq 0.3$, shaded in the figure, every protocol has a false negative rate of 1, meaning a single session never raises the alarm, yet the secure key rate is still clearly positive. At $f = 0.1$ the secure key rates are 0.322, 0.191, and

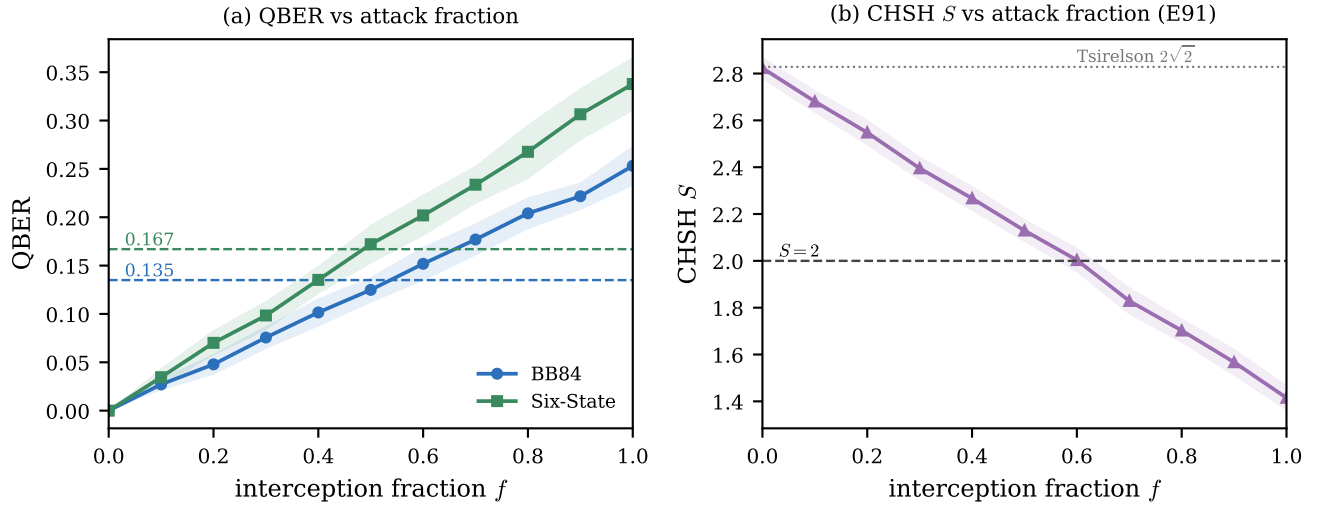


Fig. 8. Detection signal versus attack strength on an ideal channel. (a) The QBER of BB84 and the six-state protocol rises monotonically with the interception fraction f and reaches the fixed detection thresholds (0.135 and 0.167) only near $f = 0.5$. (b) The E91 CHSH value S falls from the Tsirelson bound toward the classical bound $S = 2$, which it crosses near $f = 0.6$. Shaded bands show one standard deviation over the Monte Carlo trials.

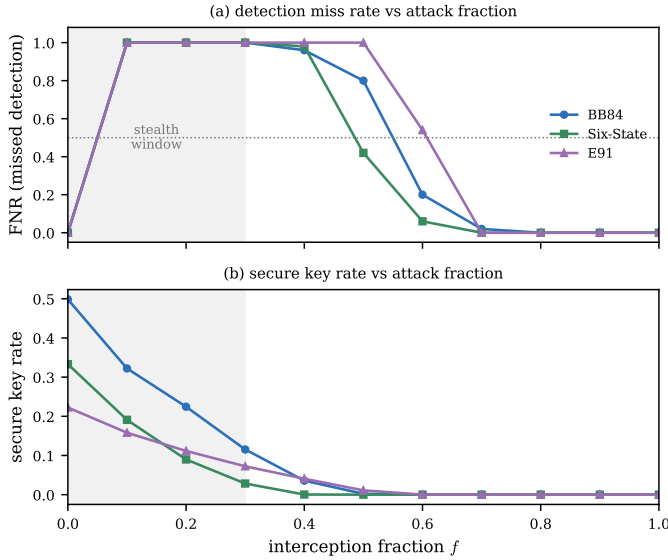


Fig. 9. The tradeoff between detection and key rate on an ideal channel. (a) The false negative rate stays at one through $f = 0.3$ (shaded stealth window) and then falls, earliest for the six-state protocol and latest for E91. (b) The secure key rate declines with f and is already near zero by the attack strength at which each protocol's false negative rate first drops below one half.

0.158 for BB84, the six state protocol, and E91 respectively, and they remain positive through $f = 0.3$. This band is a genuine stealth window, in which the attacker harvests usable key while staying invisible to per session detection. As f increases further the false negative rate falls, but the order in which the protocols become detectable matters. The six state protocol is the first to respond, with its false negative rate dropping to 0.42 at $f = 0.5$, followed by BB84, which falls to 0.20 at $f = 0.6$, and finally E91, which is still at 0.54 at $f = 0.6$ and only reaches 0 at $f = 0.7$. The six state protocol therefore closes the stealth window earliest and E91 sustains

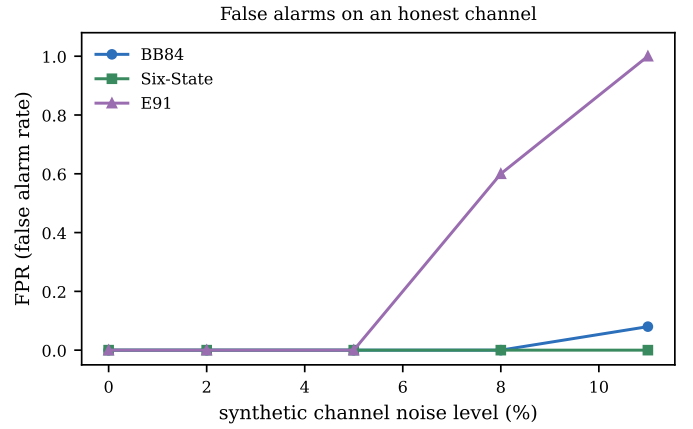


Fig. 10. False positive rate on an honest channel as the synthetic noise level increases. BB84 and the six-state protocol remain near zero, whereas the E91 CHSH test raises a false alarm on most honest sessions once the channel error rate exceeds about 8%.

it longest, which makes the entanglement based detector the easiest for a partial attacker to evade. The lower panel adds the decisive point: by the time each protocol detects the attacker reliably, its secure key rate has already collapsed to near zero. The secure key rate is below 0.01 for every protocol at the attack strength where its false negative rate first drops below 1/2. In other words, within a single session at these sample sizes, the fixed thresholds catch the attacker essentially only once the key is already destroyed, so detecting a stealthy low fraction attacker is not possible from one exchange and instead requires comparing many more signals, which we quantify in Section V-F.

D. Robustness to Channel Noise

A detector is only useful if it does not raise the alarm on an honest but noisy channel. Fig. 10 reports the false positive

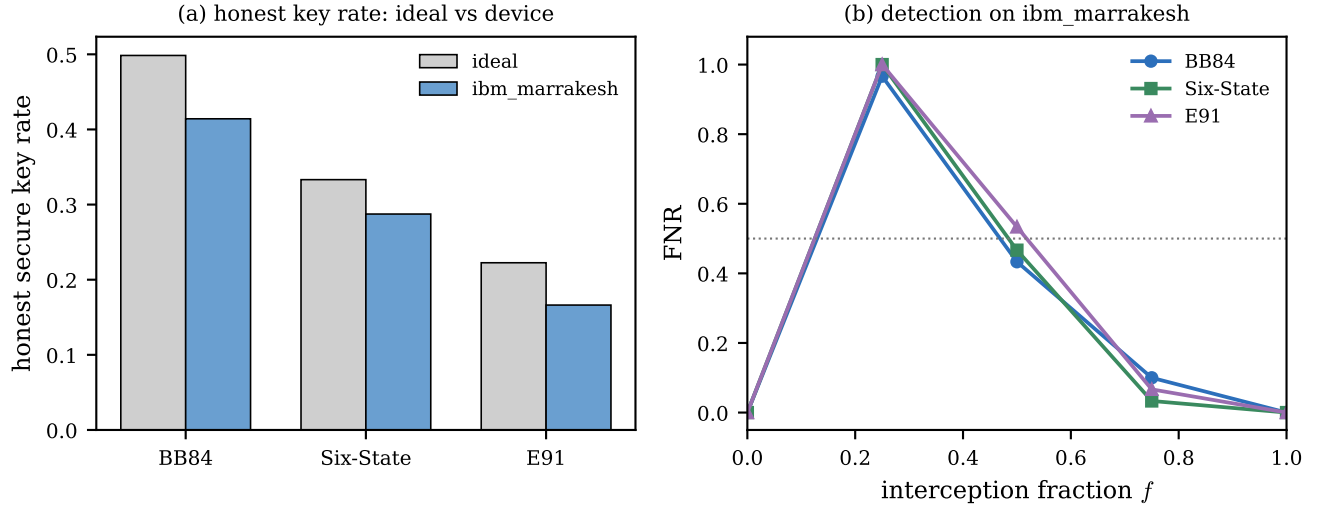


Fig. 11. Behaviour under the *ibm_marrakesh* device noise model. (a) The honest secure key rate falls only moderately relative to the ideal channel, and no protocol raises a false alarm. (b) The false negative rate versus interception fraction follows the same shape as on the ideal channel, so the stealth window persists on real hardware.

rate of each protocol as the synthetic channel noise increases with no eavesdropper present. BB84 and the six state protocol are robust: their false positive rate stays at 0 up to an honest error rate of 8%, and only BB84 shows a small rate of 0.08 at the 11% level, because its lower threshold of 0.135 sits closer to the honest error rate than the six state threshold of 0.167. E91 behaves very differently. Its honest CHSH value degrades steadily with noise, from 2.823 at 0 noise to 2.611, 2.291, 1.989, and 1.726 at the 2, 5, 8, and 11% levels, so it drops below the classical bound near the 8 % level. As a result the E91 false positive rate jumps to 0.60 at 8 % and to 1.00 at 11%, meaning the CHSH test declares an eavesdropper on essentially every honest session once the channel is moderately noisy. This fragility is the principal weakness of entanglement based detection in this study, and it reflects the fact that the CHSH value is a more delicate quantity than the QBER.

E. Behaviour Under Device Noise

The synthetic results above isolate a single noise strength, whereas a real device mixes many error sources. Fig. 11 reports the 3 protocols under the noise model extracted from the *ibm_marrakesh* backend. The honest results are reassuring. As the left panel shows, real device noise reduces the secure key rate only moderately, from 0.498 to 0.414 for BB84, from 0.333 to 0.287 for the six state protocol, and from 0.223 to 0.166 for E91, and the false positive rate is 0 for all 3. In particular the honest E91 CHSH value on the device is $S = 2.697$, comfortably above the classical bound, because the effective error rate of this backend is below the 8% level at which the synthetic CHSH test broke down. This is an important qualification of the fragility result: E91 is fragile to large channel noise, but the measured device noise is mild enough that honest entanglement based operation remains viable. The right panel shows that detection on the device follows the same pattern as on the ideal channel, with the

false negative rate staying near 1 for a 1/4 interception and falling through 1/2 near $f = 0.5$ for all 3 protocols, so the stealth window persists on real hardware.

F. Detection Resource Cost

Because a single session cannot catch a stealthy attacker, the relevant question is how many compared signals are needed to make detection statistically reliable. Fig. 12 reports the false positive rate and the false negative rate as a function of the number of compared signals K , where the false positive rate is measured on a worst case honest noisy channel and the false negative rate on a fully intercepted clean channel. To keep each protocol near its own marginal honest regime, the honest channel used for the false positive measurement is held at an 11% error rate for BB84 and the six state protocol and at a 5 % error rate for E91, since E91 at the 11% level alarms on every honest run regardless of K . The false negative rate is easy to drive down for all 3 protocols, reaching 5% or below within a few hundred compared signals. The binding constraint is the false positive rate. The six state protocol is the most sample efficient, falling below 5% by about $K = 300$, BB84 requires about $K = 1000$, and E91 lies in between, falling below 5% between $K = 500$ and $K = 1000$. All 3 reach a negligible false positive rate within a few thousand signals. The ranking mirrors the threshold geometry: the six state threshold of 0.167 leaves the widest margin above an 11% honest channel, so it suppresses false alarms with the fewest samples, whereas the BB84 threshold of 0.135 leaves the least margin and is the most sample hungry.

G. Summary of Findings

Table III consolidates the comparison across the three protocols. Taken together, the results identify the six state protocol as offering the most favourable detection profile among the 3. It produces the strongest disturbance signal

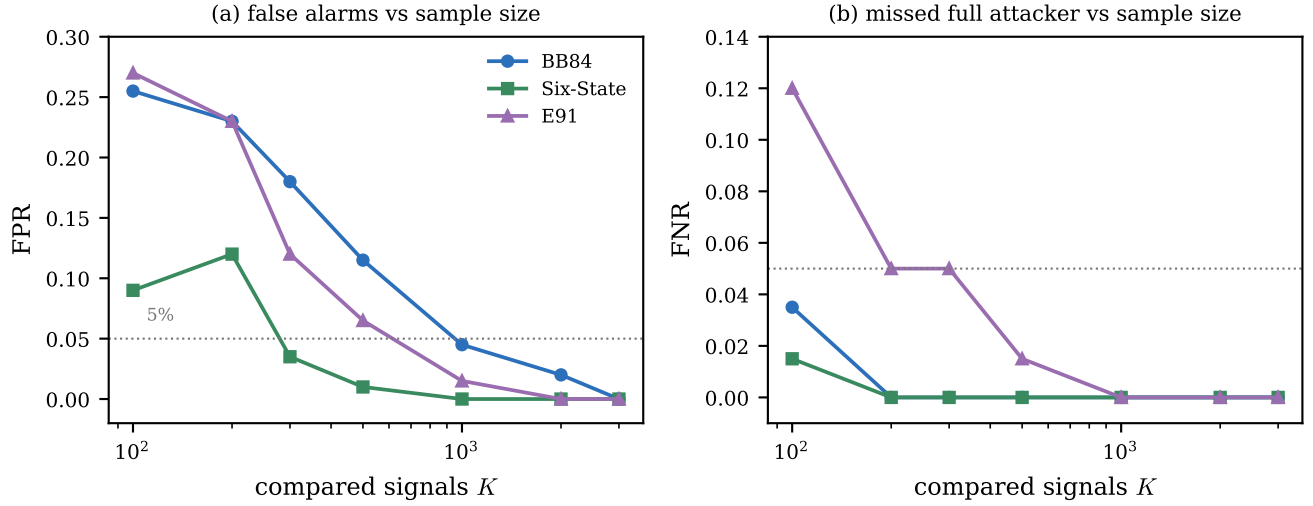


Fig. 12. Detection resource cost. (a) False positive rate versus the number of compared signals K on a worst-case honest noisy channel (11% error for BB84 and the six-state protocol, 5% for E91). (b) False negative rate versus K against a full attacker on a clean channel. The six-state protocol suppresses false alarms with the fewest samples, and the false negative rate is easy to drive down for all three protocols.

TABLE III
SUMMARY OF THE COMPARATIVE DETECTION RESULTS ACROSS THE THREE PROTOCOLS

Protocol	Detection threshold	Stealth-window edge (f)	Detection onset (f , $\text{FNR} \leq 0.5$)	FPR at 11% noise	K for $\text{FPR} \leq 5\%$	Device honest SKR
BB84	QBER 0.135	0.3	0.6	0.08	≈ 1000	0.414
Six-State	QBER 0.167	0.3	0.5	0.00	≈ 300	0.287
E91	$S = 2.0$	0.5	0.7	1.00	≈ 700	0.166

per intercepted qubit, closes the stealth window at the lowest attack fraction, never raises a false alarm across the synthetic noise range studied, and needs the fewest compared signals to make detection reliable, at the cost of the lowest sifted key yield. BB84 is intermediate on every axis. E91 directly certifies the entangled resource that secures its key, but it is the most evadable by a partial attacker and by far the most fragile to channel noise, although real device noise is mild enough that honest operation is preserved. Across all 3 protocols the central finding is the same: a partial intercept and resend attacker has a stealth window at low interception fractions in which key still flows and per session detection fails, and reliable detection of such an attacker is a question of accumulated statistics rather than of any single exchange.

VI. CONCLUSION

We presented a unified, noise aware Qiskit study of the BB84, six state, and E91 protocols that evaluates eavesdropping detection statistically, through the false positive and false negative rates, and extends this analysis from full interception to a partial intercept and resend attacker that targets a tunable fraction of the transmission. Statistical detection of this kind was established for BB84 by Lee et al. [3] and recently extended to the six state protocol by Fiorini et al. [4]; building on those thresholds, we provide the first false positive and false

negative analysis of the entanglement based CHSH detector and the first comparison of statistical detection across prepare and measure and entanglement based protocols within one framework. Across matched ideal, synthetic noise, and real device conditions, the study produces for each protocol a tradeoff between detection performance and secure key rate as a function of attack strength. The main findings are that a partial attacker has a stealth window at low interception fractions in which a usable key is still produced while a single session never raises the alarm, that reliable per session detection appears only once the attack is strong enough to have already destroyed the key, and that detecting a stealthy attacker is therefore a matter of accumulated statistics. Among the 3 protocols the six state protocol offers the most favourable detection profile, closing the stealth window earliest, never raising a false alarm across the noise range studied, and requiring the fewest compared signals, at the cost of the lowest key yield; BB84 is intermediate; and E91 directly certifies entanglement but is the most evadable and the most fragile to channel noise, although honest operation is preserved under the measured device noise.

Several directions remain open. The E91 detection boundary used here is the fixed CHSH classical bound $S = 2$, a security limit rather than a statistically optimized threshold; deriving the threshold on S that balances the false positive and false

negative rates from the variance of the CHSH estimator, in the way Lee et al. [3] and Fiorini et al. [4] do for the QBER, would give the entanglement based detector the same statistical footing as the prepare and measure detectors. The six state detection threshold was placed by balancing the Hoeffding bounds on the two detection errors; a tighter construction that instead minimizes their weighted sum, together with distribution aware rather than worst case bounds, would refine it further. The secure key rate used the asymptotic Shor and Preskill expression, and a finite key treatment would tighten the resource cost results. Because the central finding is that detection is a matter of accumulated statistics, a sequential detector that aggregates evidence across sessions is a natural way to catch a low fraction attacker that no single session catches. Finally, executing the protocols on real quantum hardware rather than a device noise model, and extending the adversary beyond intercept and resend to collective and coherent attacks, would test whether the stealth window and the protocol ordering observed here persist under stronger conditions.

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REFERENCES

- [1] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," in *Proc. IEEE Int. Conf. Computers, Systems and Signal Processing*, pp. 175–179, 1984.
- [2] A. K. Ekert, "Quantum cryptography based on bell's theorem," *Physical Review Letters*, vol. 67, no. 6, pp. 661–663, 1991.
- [3] C. Lee, I. Sohn, and W. Lee, "Eavesdropping detection in BB84 quantum key distribution protocols," *IEEE Transactions on Network and Service Management*, vol. 19, no. 3, pp. 2689–2701, 2022.
- [4] F. Fiorini, R. G. Garroppo, M. Pagano, and R. Schiavini Yadzhak, "Eavesdropper detection in six-state protocol against partial intercept-resend attack," *Future Internet*, vol. 17, no. 11, p. 500, 2025.
- [5] N. U. Ain, M. Waqar, A. Bilal, A. Kim, H. Ali, U. U. Tariq, and M. S. Nadeem, "A novel approach based on quantum key distribution using BB84 and E91 protocol for resilient encryption and eavesdropper detection," *IEEE Access*, vol. 13, pp. 32819–32833, 2025.
- [6] P. Dhakal, B. R. Dawadi, and N. B. Adhikari, "Performance analysis of different quantum key distribution protocols for optimised security and efficiency," *IET Quantum Communication*, p. e70015, 2025.
- [7] M. H. Saeed, H. Sattar, M. H. Durad, and Z. Haider, "Implementation of QKD BB84 protocol in Qiskit," in *Proc. 19th Int. Bhurban Conf. Applied Sciences and Technology (IBCAST)*, pp. 689–695, 2022.
- [8] N. M. Kadir and M. A. Hossain, "Simulation of E91 quantum key distribution protocol using Qiskit," in *Proc. IEEE 2nd Int. Conf. Computing, Applications and Systems (COMPAS)*, (Kushtia, Bangladesh), pp. 1–6, 2025.
- [9] L. Mariani, L. Salatino, C. Attanasio, S. Pagano, and R. Citro, "Simulation of an entanglement-based quantum key distribution protocol," *The European Physical Journal Plus*, vol. 139, no. 7, p. 602, 2024.
- [10] H. Bechmann-Pasquinucci and N. Gisin, "Incoherent and coherent eavesdropping in the six-state protocol of quantum cryptography," *Physical Review A*, vol. 59, no. 6, pp. 4238–4248, 1999.
- [11] J. Singh, S. Ghosh, Arvind, and S. K. Goyal, "Role of Bell-CHSH violation and local filtering in quantum key distribution," *Physics Letters A*, vol. 392, p. 127158, 2021.
- [12] S. DiAdamo, J. Nötzel, B. Zanger, and M. M. Beşe, "QuNetSim: A software framework for quantum networks," *IEEE Transactions on Quantum Engineering*, vol. 2, pp. 1–12, 2021.
- [13] F. Fiorini, M. Pagano, R. G. Garroppo, and A. Osele, "Estimating interception density in the BB84 protocol: A study with a noisy quantum simulator," *Future Internet*, vol. 16, no. 8, p. 275, 2024.
- [14] F. Fiorini, R. G. Garroppo, and M. Pagano, "Threshold-based eavesdropper detection for partial intercept-resend attack in noisy BB84 quantum key distribution," *Computer Networks*, vol. 277, p. 112058, 2026.
- [15] S. M. Badiehzadegan, "BB84 quantum key distribution simulation," 2026. Independent researcher. Available: https://github.com/mbzadegan/PhD_Research.
- [16] Y.-Y. Fei, X.-D. Meng, M. Gao, H. Wang, and Z. Ma, "Quantum man-in-the-middle attack on the calibration process of quantum key distribution," *Scientific Reports*, vol. 8, no. 1, p. 4283, 2018.
- [17] B. Li, S. Ahmed, S. Saraogi, N. Lambert, F. Nori, A. Pitchford, and N. Shammah, "Pulse-level noisy quantum circuits with QuTiP," *Quantum*, vol. 6, p. 630, 2022.
- [18] P. Zeng, D. Bandyopadhyay, J. A. Méndez Méndez, N. Bitner, A. Kolar, M. T. Solomon, F. J. Heremans, D. D. Awschalom, L. Jiang, and J. Liu, "Towards efficient and secure quantum-classical communication networks," in *Proc. IEEE 6th Int. Conf. Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA)*, pp. 520–523, 2024.